Probability, Lecture 3-4, reminder

Quantities governed by randomness correspond to functions on the probability space called random variables. The value taken by a random variable is subject to chance, and the associated likelihoods are described by a function called the distribution function. Two important classes of random variables are discussed, namely discrete variables and continuous variables. The law of averages, known also as the law of large numbers, states that the proportion of successes in a long run of independent trials converges to the probability of success in any one trial. This result provides a mathematical basis for a phylosophical view of probability based on repeated experimentation. Worked examples involving random variables and their distributions are included.

We can only speak about "probability" if we have some experience (or presumptions) about the subject. Either we can make logical deductions (like in the case of coins or dice), or we must make observations first. That means, we repeat a relevant experiment n times, each experiment shall be identical in its conditions and independent of the others (in the simplest case), and we draw conclusions about the subject, based on the sequence of the outcomes. Mathematically speaking, the experiments correspond to random variables $X_1, X_2, ..., X_n$, and the outcomes of the experiments can be described by real numbers $x_1, x_2, ..., x_n$ − our observations, a *sample*.

Statistics is the science dealing with observed data. For summarising data, the basic tools of statistics are as follows.

Three different *measures of location* can be used to describe the *centre of a set of data*: mode, median and mean.

The *mode* is the value that occurs most often. This is used when the data is qualitative, or quantitative with either a single mode or two modes (bimodal). Not informative if each value occurs only once.

The *median* is the middle value when the data is put in order (or the average of two values in the middle, if the number of elements is even). This is used for quantitative data. It is usually used when there are a few extreme values or errors, because these do not affect the median.

The *mean* is the sum of all observations divided by the total number of observations − just an other name for the arithmetical *average*, notation: $\bar{x}$. This is used for quantitative data and uses all the pieces of data. It therefore gives a true measure of data. However, it is affected by extreme values and data errors.

The *measures of dispersion* are: range, interquartile range, variance and standard deviation (the latter two are measuring the deviations from the centre).

Range = highest value − lowest value.

Quartiles split the data into four parts: exactly 25% of the observations should have a value less than the lower quartile (Q_1), 50% of the observations should have a value below the median (Q_2), and 75% of the observations should have a value below the upper quartile (Q_3).

Interquartile range $= Q_3 - Q_1$.

Variance: The deviation of an observation (x_i) from the mean ($\bar{x}$) is given by $x_i - \bar{x}$ (for $i = 1, 2, ..., n$). In order to make all these differences positive, take their squares, and then the average of the squares. So one way of measuring the total dispersion of a set of data is to use the variance:

$$\sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n}.$$

Basic calculation yields an other version of this formula: $\sum \frac{(x_i-\bar{x})^2}{n} = \frac{\sum x_i^2}{n} - (\frac{\sum x_i}{n})^2$.

The *standard deviation* (SD) is the square root of the variance. That is, the quadratic average of the deviations from the mean.

Example: for the data $1, 2, 2, 3, 4, 5, 5, 6, 7, 8, 9, 9, 9, 10, 11$ (after ordering), the mode is 9, the median is 6, the mean is 6.067. Range is 11 - 1 = 10, the quartiles are 3, 6 and 9, the interquartile range is 9 - 3 = 6. The variance is 9.662, the standard deviation is $\sqrt{9.662} = 3.108$ (check these numbers, using a spreadsheet, e.g. Excel).

Other example: for the data $1, 2, 3, 4, 5, 6$, the mean is 3.5. The deviations from the mean are -2.5, -1.5, -0.5, 0.5, 1.5, 2.5. The variance is $\frac{(2.5^2+1.5^2+0.5^2)\times 2}{6} = 2.917$, the standard deviation is $\sqrt{2.917} = 1.708$.

After collecting a lot of data, we can pict the values, making a histogram, to see how they are distributed on the real line. In some cases, the same values turn up many times, and we can form a concept about their probability based on their relative frequency. These are called *discrete distributions.*

In these cases (e.g. playing with dice or cards), the random variable in question (X) has finitely many or countably many possible values only, and the probabilities $P(X = x)$ for some x's ($x \in \mathbb{R}$ in general, but usually $x \in \mathbb{N}$) are sufficient to define the distribution.

In other cases, there are no repeated values, just values close to each other, because any outcome is possible in an interval at least, on a continuous scale. Like when you are measuring the hight of students in a class: if you made exact measurements with great precision, you wouldn't find two students with equal hight. These are called *continuous distributions.* In such a case, you can't make a histogram based on repeated values, but you must partition the range to sub-intervals, and count the values falling into each sub-interval, and plot these counts. You can decide what partitioning you choose for the plotting, but it's better to make the sub-intervals equally sized. You can go down with the size of the intervals, as long as there are enough values falling in each one. Then you can imagine a (mostly continuous) function that is *proportional* to the number of cases falling in the sub-intervals (if you chose equal-length subintervals). More precisely, the area under the curve of this function in a sub-interval, should be proportional to the number of cases falling there (this latter statement works for non-equal subintervals, too). This argument leads to the concept of *probability density function.*

An easy example is when we (uniformly) randomly chose a point (X) in the [0,1] interval. Choosing a pre-given number exactly (like $X = 1/\pi$ or anything between 0 and 1) has probability zero (if we are not inclined to choose short decimal numbers). But this is "not the same zero" as the probability of choosing $X = 2$ (which is impossible). In this simple example, the density function will be constant 1 over the interval [0,1], and zero everywhere else.

The integral of the density function over any given interval (= the area under the graph of the density function) must give the probability of X falling into that interval. The density function must be non-negative, and its integral on the whole real line must be 1 (the total probability). Changing it in a few points does not matter because it won't change the integral.

This may be enough for an intuitive understanding of probability density functions, but to gain an exact definition, we must define the (cumulative) distribution function first.

The *cumulative distribution function* (CDF) of a random variable X is defined as a probability:

$$F_X(x) = P(X \leq x),$$

where x (the argument of the function) can be any real number. This gives a unified way to describe any distribution on the real numbers, of any type. Let's see an example.

A dart is flung at a circular target of radius 3. We can think of the hitting point as the outcome of a random experiment.

a) We shall suppose for simplicity that the player is guaranteed to hit the target somewhere. Let us suppose that the probability that the dart lands in some region A is proportional to its area $|A|$. The target is partitioned by three circles C_1, C_2, C_3, centred at the origin, with radius 1, 2, 3 respectively, and the scoring system is based on these circles: the score (X) is the upper integer part of the distance of the dart from the origin.

Now X is a purely discrete random variable: it attains values $0, 1, 2, 3$ with probability $0, \frac{1}{9}, \frac{4-1}{9}, \frac{9-4}{9}$ respectively. The cumulative distribution function of X will be discontinuous at the points $1, 2, 3$, and constant in between. E.g. $P(X \leq 2.5) = P(X \leq 2.7) = P(X \leq 2) = \frac{4}{9}$. In general, $F_X(r) = P(X \leq r)$ equals 0 for $r < 1$, equals $\frac{1}{9}\lfloor r \rfloor^2$ for $1 \leq r < 3$, and equals 1 for $3 \leq r$.

b) Let us consider a revised method of scoring: the player scores an amount (Y) equal to the distance between the hitting point and the centre of the target.

This time $F_Y(r) = P(Y \leq r)$ is continuous. $F_Y(r) = \frac{1}{9}r^2$ when $0 \leq r < 3$ (and again 0 for $r < 0$, and 1 for $3 \leq r$).

c) Now suppose that the player fails to hit the target with probability p. If he is successful then he scores an amount (Z) equal to the distance between the hitting point and the centre of the target; if he misses then he scores 4.

This is a mixed case: the distribution of Z is neither discrete nor continuous. See page 37 in the Grimmet-Stirzaker book (Probability and Random Processes, uploaded to Canvas) for the graphs of the CDF's F_X, F_Y, F_Z.

Note that a CDF must always be monotonically increasing; also note that in $-\infty$ the CDF must tend to 0 and in $+\infty$ it must tend to 1. (In this setup, the CDF is right-continuous. Some sources define if with $<$ instead of $\leq$, which makes it left-continuous.)

The derivative of a monotonic function is defined everywhere on the real line except for countably many points. When the cumulative distribution function is continuous, its derivative (its rate of change) is the *probability density function* (PDF). This is exactly the case of continuous distributions, discussed above.

When the distribution is discrete, the CDF is a step function (see case a).

We defined above the mean, variance and standard deviation of *samples.* When we have a picture about the distribution behind the finite set of data, then we can make inferences about the result of further experiments. For example, if we roll a die, $1, 2, 3, 4, 5, 6$ are the possible outcomes shown on the upper face, and they are equally likely if the die is fair, regular. The distribution of the face-up number (X) is: $P(X = k) = \frac{1}{6}$ for $k = 1, 2, ..., 6$. Even if the first 6 numbers thrown are not these 6 numbers each occurring exactly once, we know that *in the long run* we will get all these numbers with (approximately) equal frequency (this is the Law of Large Numbers, see lesson 7-8). The *expectation* or *expected value* is the average of all possible values, weighted by their theoretical probabilities. We denote it by EX:

$$EX = \sum k \times P(X = k),$$

which equals $\sum_{k=1}^{6} k \times \frac{1}{6} = 3.5$ in the current example.

Similarly, we can define the theoretical variance and standard deviation of a *distribution* (more precisely, the variance and standard deviation of a random variable, having the given distribution). As we calculated arithmetical average of the squared deviations from the sample mean, here we calculate expected value of the squared deviations from the theoretical mean:

$$Var(X) = E\{[X - E(X)]^2\} = E(X^2) - [E(X)]^2$$

(the second part will be proven in Lecture notes 5-6).

Now let's write down the most important *discrete distributions.* The first two are quite natural. The third one is more abstract, but it follows from the second one, in continuous time.

1) *Geometric* distribution with parameter p.

Consider an experiment, where the possible outcomes are 1 (success) and 0 (failure). The probability of success at each experiment is p. You repeat this experiment, independently of one another, until the first success. Let X be the number of experiments necessary. Then

$$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, ...$$

Calculations yield $E(X) = \frac{1}{p}$, $Var(X) = \frac{1-p}{p^2}$, $SD(X) = \frac{\sqrt{1-p}}{p}$.

An example is to roll a die until we get a six. The number of necessary rolls follows a Geometric distribution with parameter $p = \frac{1}{6}$. So e.g. $P(\text{the first 6 comes at the 5th roll}) = \left(\frac{5}{6}\right)^4 \left(\frac{1}{6}\right) = 0.0804$. The expected value of the number of necessary rolls is $\frac{1}{p} = \frac{1}{\frac{1}{6}} = 6$, the standard deviation is $\frac{\sqrt{1-p}}{p} = \frac{\sqrt{1-\frac{1}{6}}}{\frac{1}{6}} =$ 5.477.

2) *Binomial* distribution with parameters n and p.

Consider an experiment, where the possible outcomes are 1 (success) and 0 (failure). The probability of success at each experiment is p. You repeat this experiment n times, independently of one another. Let Y be the number of successful experiments. Then

$$P(Y = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, 2, ..., n.$$

Calculations yield $E(Y) = np$, $Var(Y) = np(1-p)$, $SD(Y) = \sqrt{np(1-p)}$.

An example is to roll a die $n = 6$ times. Here the number of sixes follows a Binomial distribution with parameters $n = 6$ and $p = \frac{1}{6}$. So e.g. P(2 sixes out of 6 rolls)$= \binom{6}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^4 = 0.201$. The expected value of the number of sixes is $np = \frac{6}{6} = 1$, the standard deviation is $\sqrt{np(1-p)} = \sqrt{1 \times \frac{5}{6}} = 0.913$.

3)*Poisson* distribution with parameter λ.

Let Z be the number of events (belonging to a specific type) occurring in a given time period, $-$ given the *average* number of times the event usually occurs over that time period (this is the parameter, λ). For example, number of phone calls coming in to a call centre in a 5 minute period, $-$ knowing the average number of phone calls coming in. Then

$$P(Z = k) = \frac{\lambda^k}{k!}e^{-\lambda}, \quad k = 0, 1, 2, ...,$$

and $E(X) = \lambda$, $Var(X) = \lambda$, $SD(X) = \sqrt{\lambda}$.

For example if we know that the average number of phone calls coming in to our call center in a five minute period is 10, then we have a Poisson distribution with parameter 10 when we consider the number of phone calls coming in between 1 pm and 1:05 pm today. That means P(no phone calls in this period)$=$ $\frac{10^0}{0!}e^{-10} = \frac{1}{1}e^{-10}$ which is very small. The expected value is 5, and the standard deviation is $\sqrt{5} = 2.236$.

Remark: We can understand the Poisson distribution by dividing the time interval in the above description of the Poisson distribution to n equal parts, and repeating an experiment in each interval, with success rate p, such that $np = \lambda$. The number of successful experiments is $Y \sim Binomial(n, p)$. Then divide each interval to two equal parts, and make the probability of success half of the previous success rate. And so on, n shall tend to infinity and the probability of success in each interval, p, shall tend to zero, keeping $np = \lambda$ constant. This yields the given formula for $P(Z = k)$ in the Poisson distribution as the limit of the probabilities $P(Y = k)$ in the Binomial distribution.

Consequently, if n is very big and p is very small in the Binomial distribution, it can be approximated by a Poisson distribution where $\lambda = np$.

The Binomial distribution is to be used when
a) there is a fixed number of trials (n);
b) the trials must be independent;
c) there are only two possible outcomes: success or failure;
d) the probability of success (p) is constant for each trial.
Further examples of Binomial distribution:
- The number of sixes obtained, when a fair cubical die is rolled n times.
- The number of girls in a family of 4 children.

The Poisson distribution is to be used when
a) events occur independently of each other;
b) in continuous space or time;
c) at a constant rate: mean number in an interval is proportional to the length of the interval.
They are called "random" or "rare" events.
Further examples of Poisson distribution:
- Number of radioactive particles being emitted by a certain source during a 5 minute period.
- Number of red cars passing your college gates in a 10 minute period.